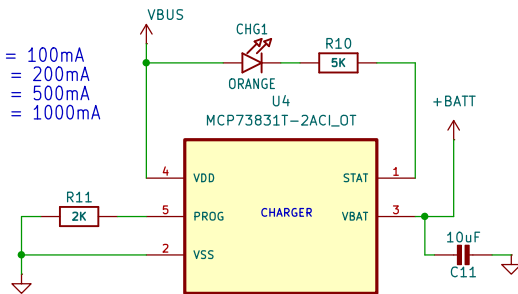


10K	=	100mA
5.0K	=	200mA
2.0K	=	500mA
1.0K	=	1000mA



POWER AND FILTERING

VBUS

+BATT

R12 100K

D1 MBR0540

Q1 DMG3415U

R9 100K

JST_ONOFF

BATT1 JST_PH

+3.3V

U3 RT9080-33GJ5

C9 0.1uF

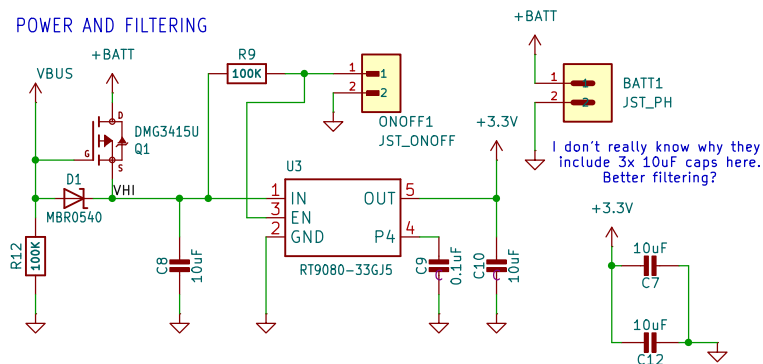
C10 10uF

C7 10uF

C12 10uF

I don't really know why they include 3x 10uF caps here. Better filtering?

This is copied from the Adafruit ESP32-Feather schematic
<https://github.com/adafruit/Adafruit-ESP32-Feather-V2-PCB>



U5
MAX17048G_T10

CTG 1
CELL 2
VDD 3
EP 9
ALERT 5
GND 4

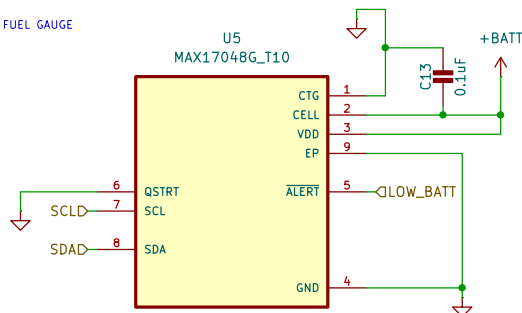
QSTRT 6
SCL 7
SDA 8

C13 0.1uF

+BATT

LOW_BATT

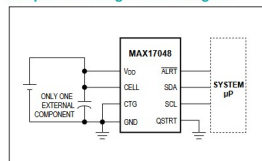
For this model of fuel gauge, VDD connects to the BAT line. That feels cursed, but it's what the datasheet says



For this model of fuel gauge,
VDD connects to the BAT line.
That feels cursed, but it's
what the datasheet says

The diagram shows the MAX17048 chip with the following connections:

- V_{DD}** is connected to a positive supply rail.
- CELL** is connected to a battery symbol, with a note: "ONLY ONE EXTERNAL COMPONENT".
- CTO** is connected to ground.
- GND** is connected to ground.
- AUT** is connected to the **SYSTEM μ P**.
- SDA** is connected to the **SYSTEM μ P**.
- SCL** is connected to the **SYSTEM μ P**.
- QSTR** is connected to ground.



PMBUMP		NAME	FUNCTION
TDFN	WLP		
1	A1	CTG	Connect to Ground
2	A2	CELL	Connect to the Positive Battery Terminal. MAX17046: Not internally connected. MAX17049: Voltage sense input.
3	A3	V _{DD}	Power-Supply Input. Bypass with 0.1µF to GND. MAX17046: Voltage sense input. Connect to positive battery terminal. MAX17049: Connect to regulated power-supply voltage.
4	A4	GND	Ground. Connect to negative battery terminal.
5	B4	ALRT	Open-Drain, Active-Low Alert Output. Optionally connect to interrupt input of the system microcontroller.
6	B3	QSTRT	Quick-Start Input. Allows reset of the device through hardware. Connect to GND if not used.
7	B2	SCL	I ² C Clock Input. SCL has an internal pulldown (ppp) for sensing disconnection.
8	B1	SDA	Open-Drain I ² C Data Input/Output. SDA has an internal pulldown (ppp) for sensing disconnection.
—	—	EP	Exposed Pad (TDFN Only). Connect to GND.

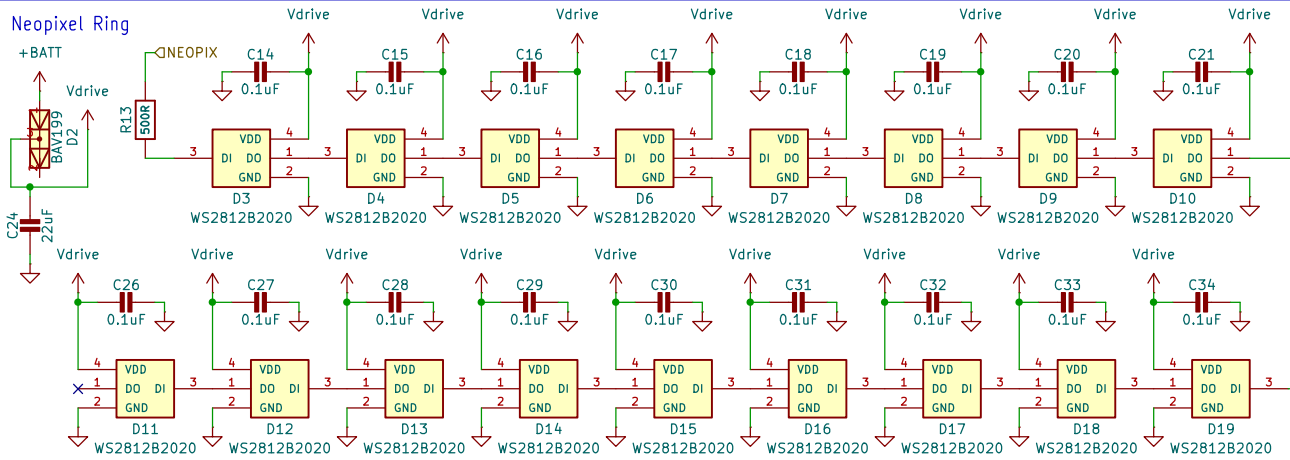
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Title: *Smart Compass*

SIZE: A1	1
KiCad E.D.A. 10.0.1	

Id: 2/4

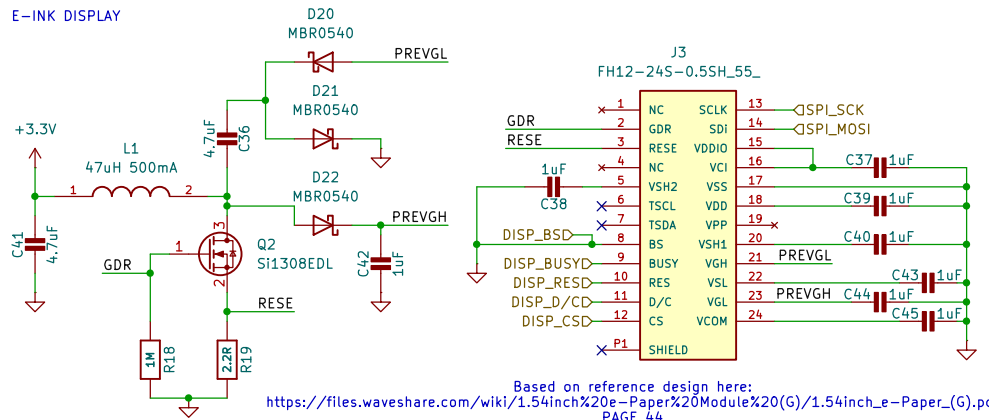
Neopixel Ring



No more than 4 pixels will be driven at full brightness, or 8 at 1/2 brightness, at a time.

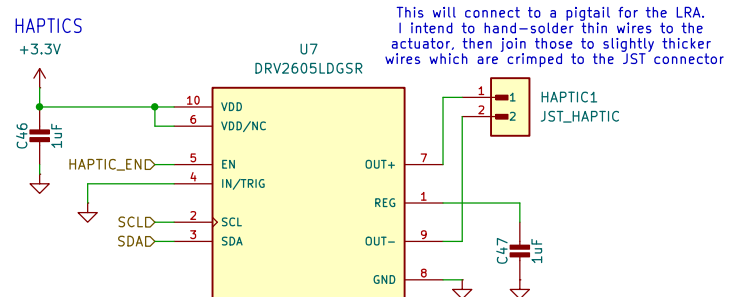
Adafruit estimate 20mA for typical use
20mA * 4 = 80mA LED load

E-INK DISPLAY



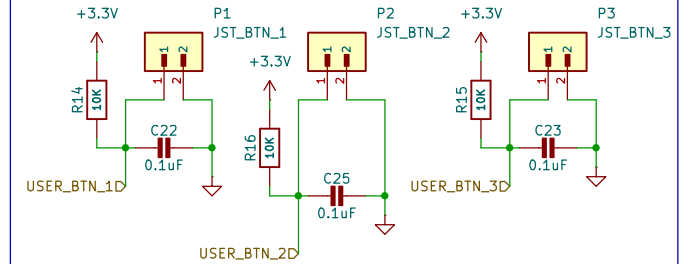
Based on reference design here:
[https://files.waveshare.com/wiki/1.54inch%20e-Paper%20Module%20\(G\)/1.54inch_e-Paper_\(G\).pdf](https://files.waveshare.com/wiki/1.54inch%20e-Paper%20Module%20(G)/1.54inch_e-Paper_(G).pdf)
PAGE 44

HAPTICS

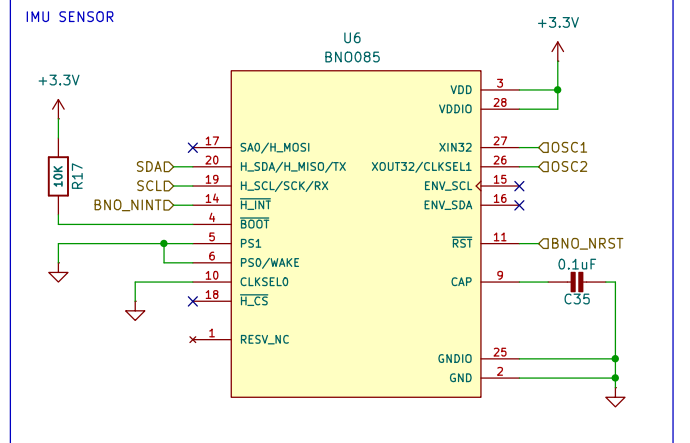


This will connect to a pigtail for the LRA. I intend to hand-solder thin wires to the actuator, then join those to slightly thicker wires which are crimped to the JST connector

USER BUTTONS



IMU SENSOR



Sheet: /User IO/
File: user_io.kicad_sch

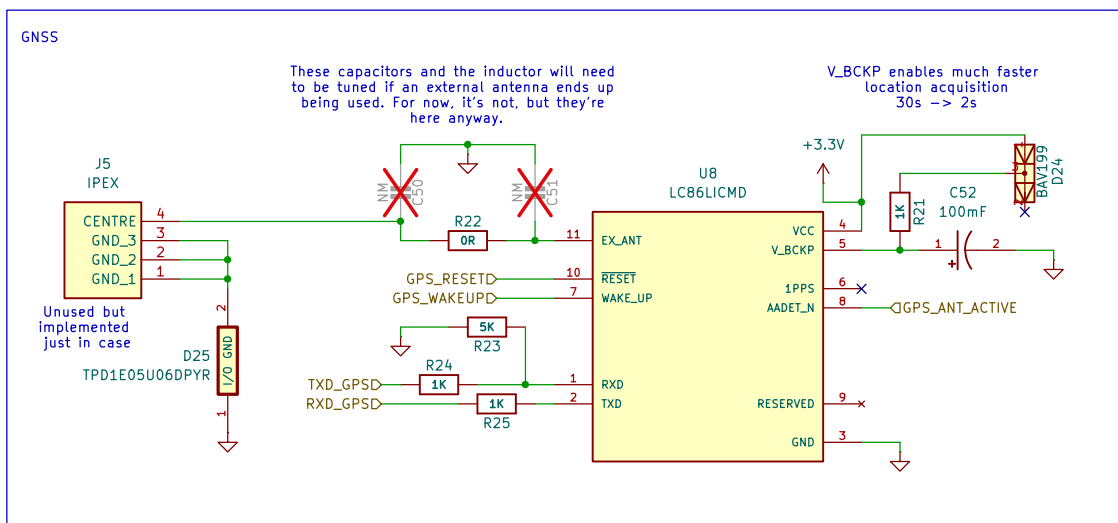
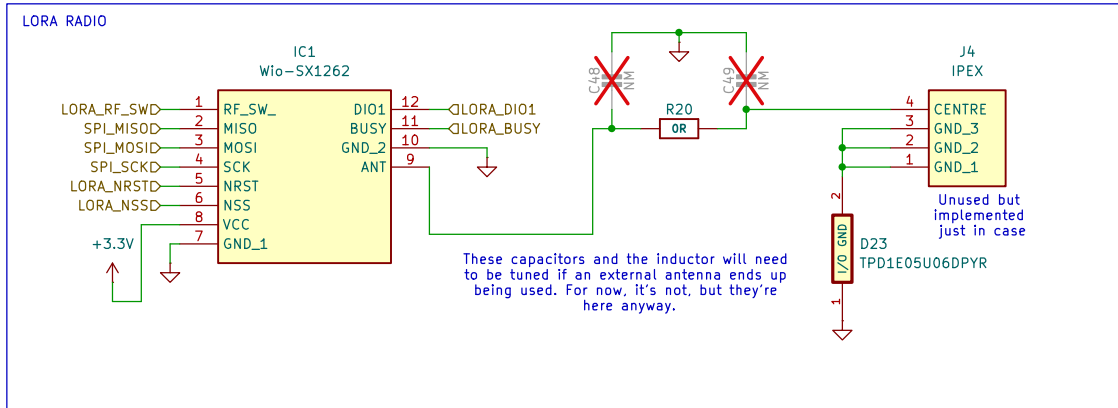
Title: Smart Compass

Size: A4 Date: 2026-06-04

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Rev: 0.2

Id: 3/4



Sheet: /Communications/
File: communications.kicad_sch

Title: Smart Compass

Size: A4 Date: 2026-06-04
KiCad E.D.A. 10.0.1

Rev: 0.2
Id: 4/4